MIL-L-2668D
3 December 1969
SUPERSEDING
MIL-L-2668C
15 August 1968

MILITARY SPECIFICATION

LIGATURE, UMBILICAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a sterile umbilical ligature (tape) suitable for tying the umbilical cord of newly born infants.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive, Water Resistant, For Sealing Fiberboard Boxes.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing).

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6515

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

UNITED STATES PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States (U.S.P.).

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.

3. REQUIREMENTS

3.1 Material.

3.1.1 Cotton. The ligature (tape) shall be made from bleached cotton with a diagonal weave, over two and under two, from border to border. The finished tape shall have the physical properties indicated in Table I, when tested as specified in 4.4.1 and 4.4.2.

Table I. Physical Properties.

Thread	2-ply
Number of threads in tape width (1/8 inch)	17
Breaking strength of tape	25-pound (minimum)

3.1.2 Plastic (container). Plastic shall be of good quality, and of proper permeability, to be suitable for the purpose intended. The plastic shall be nontoxic, and compatible with the contents of the container.

3.1.3 Aluminum foil (container). The aluminum foil shall be of good quality, suitable for the purpose intended.

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3.1.4 Reels. Reels shall be fabricated from suitable nontoxic material that is compatible with the contents of the container.

3.2 Description. The umbilical ligature shall consist of a cotton tape as specified in 3.1.1. The tape shall be 1/8 ± 1/64 inch in width and 18 inches (minimum) in length. The ligature shall be suitable for use in tying the umbilical cord of newborn infants. The ligature shall be sterile, evenly coiled or wound on a reel in such a manner as to permit ready dispensing, and shall be sealed in an immediate container as specified in 3.2.1.

* 3.2.1 Immediate container. The specified sterile ligature shall be placed in a double container. The completed, sealed, individual package shall consist of the sterile ligature in the inner container enclosed in turn in an outer container which shall maintain the inner container sterile, through changing temperatures from minus 25°C. to 40°C. The package shall be so constructed as to be readily opened. The double container shall consist of an inner and outer container fabricated of plastic, aluminum foil, or combination thereof (see 3.1.2 and 3.1.3). The ligature shall be placed in the inner container, without packet fluid. Closure of the inner container shall be accomplished by heat sealing. The container shall be fabricated in such a manner, and of such material, as to maintain the sterility of the ligature. This container in turn, shall be placed in an outer container fabricated of plastic or foil as specified above. Closure of the outer container shall be accomplished by heat sealing. The outer container shall be so designed as to permit ready dispensing of the contents by sterile techniques and without damaging the inner container. The sealed outer container design shall include two free tabs along the top edge. The tabs shall be capable of being readily peeled back along the sealed edges to expose the inner container as shown in figure 1. This shall be done without contaminating the inner container when opening the outer container and removing the inner container. The outer container shall be fabricated in such a manner, and of such material, as to maintain the sterility of the inner container. The inner and outer container shall be tested for sterility and leakage as specified in 4.4.3. In addition, the inner container, when tested as specified in 4.4.4, shall not burst, or leak (indicated by a steady stream of gas bubbles). The inner containers shall be capable of being resterilized in a solution of 98.5 percent alcohol and 1 percent formaldehyde for 30 days without its deterioration and without any harmful effects on the ligature contained therein. When tested as specified in 4.4.5, the outer container shall not leak (including by capillary action) so as to affect the sterility of its contents. The outer container need not

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comply with this leakage test if 12 ligatures in their immediate containers, are heat sealed in a 4 mil polyethylene bag prior to their insertion into the required unit package.

3.3 Sterility. The contents of the outer container (including the inner container and the contents of the inner container (including ligature, reel and fluid) shall be sterile. The ligature and the inner container shall comply with requirements for sterility as contained in the current edition of the U.S.P., when tested as specified in 4.4.3.

3.4 Identification marking. Each immediate container shall be provided with a commercial label identifying the ligature as to its composition, width, length, name of contractor, the legend "sterile", the legend "nonoilable" and other requirements of statute. As an alternate, the registered trademark may be substituted for the name of the contractor. Reels may be used as labels. The label on the inner container shall be visible through the outer containers.

3.5 Control (lot) numbers. The manufacturer shall maintain control (lot) numbers in order to identify each lot. The control (lot) numbers shall be entered in the manufacturer's original and permanent records of sterilization and shall also appear on each shipment, as hereinafter specified. The manufacturer's records shall be open for Government inspection. When sterilization is accomplished in two steps, only one control (lot) number need be indicated. However, the manufacturer shall maintain records which will identify the contents with both sterilization lots.

3.6 Certificate of sterility. A certificate of sterility, listing all lot or control numbers shall state that samples from each sterility lot or control have been tested in accordance with United States Pharmacopeia (U.S.P.) method and have been found to be sterile. The certificate shall reference the method used in determining sterility. For destination inspection only, one copy of a certificate of sterility shall be submitted by the contractor directly to Defense Personnel Support Center, Directorate of Medical Material, 2800 South 20th Street, Philadelphia, Pa. 19101, attention: Quality Control Branch, and one copy shall be forwarded with each shipment to the consignee. For source inspection, one certificate shall be furnished to the procuring activity via the government inspector.

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3.7 Instructions. Each unit package shall be provided with instructions describing the proper techniques for dispensing the sterile ligatures from the containers. In addition, information cautioning the user against the use of chemicals, anticorrosive agents and solutions that would be harmful to the inner container shall be included, along with a statement (when applicable) indicating the maximum number of days that inner containers may remain in sterilization solutions without harm to the ligatures.

3.8 Workmanship. Workmanship shall be first class throughout. Ligatures shall be smooth, free from kinks, splinters, fraying, unraveling, cuts or knots. The entire assembly, including ligatures, reels, fluid and container, shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation with the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

* 4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the cotton, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling for lot acceptance.

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4.2.1 Inspection lot. Except where otherwise indicated, the inspection lot shall be determined in accordance with Military Standards MIL-STD-105, and shall be based on the unit of product being one ligature (end item).

4.2.2 Sampling for examination. A random sample of ligatures shall be selected from each lot in accordance with Military Standard MIL-STD-105. The inspection level and Acceptable Quality Level (AQL) shall be as indicated in Table II.

Table II. Sampling for Examination

Type defect	Inspection level	AQL (Percent defective)
Major A	II	1.0
Major B	S2*	1.0
Minor	II	2.5

*If acceptance number is not exceeded, lot shall be acceptable. If rejection number is exceeded, inspection at level II may be performed to determine final rejection or acceptance.

4.2.3 For tests. Sampling for tests, shall be conducted as specified in Table III. Tests for the ligature proper shall be performed on the finished sterilized ligature.

Table III. Sampling for Tests 1/

Component	Characteristic	Rqmt.	Test procedure	Inspection level	Acceptance number
Ligature and container	Sterility and leakage	3.2.1 & 3.3	4.4.3	2/	0
Ligature	Breaking strength	3.1.1	4.4.1	3/	0
Inner container	Container leakage test	3.2.1	4.4.4	S-1 (4/)	0
Outer container	Container leakage test	3.2.1	4.4.5	S-1	0

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1/ Except as otherwise indicated, sampling shall be conducted as specified in Military Standard MIL-STD-105.

2/ Sampling for each test shall be conducted in accordance with the current edition of the U.S.P.

3/ Five units shall be selected at random from each lot for testing.

4/ The inspection lot for container leakage test shall not exceed 20,000 containers, or one day's uninterrupted production.

4.2.4 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The ligature shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in Tables II and IV. Inspection, however, shall not be restricted to defects listed in Table IV. Any ligatures in the sample containing one or more defects shall be rejected, and if the number of defective ligatures in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table IV. Classification of Defects in Accordance with Military Standard MIL-STD-105

Categories	Defects
Critical	None defined.
Major A	
101	Holes or tears in inner or outer container.
102	Opening in seal of inner or outer container.
103	Required label for inner container not visible, incorrect, not legible, incomplete, or missing.
Major B	
151	Immediate container does not readily, or properly, open.
152	Ligature not flexible.
153	Surface of ligature not smooth and free of fraying, unraveling kinks, cuts, or knots.
154	Ligature not one continuous piece, of length, width, ply, and number of threads in tape width, as specified.
155	Ligature and contents of inner and outer container not free of not free of foreign matter or discoloration.
156	Ligature entangled, snarled, knotted, or other defects which prevent its ready dispensing from reel or coil.

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Table IV. Classification of Defects in Accordance with Military Standard MIL-STD-105 (cont'd.)

Categories	Defects
Minor	
201	Inner container (aluminum foil only), not marked to indicate proper point for opening.
202	Instructions incorrect, illegible, incomplete, or missing.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements

4.4.1 Breaking strength. The breaking strength shall be determined by testing on standard tensile strength tester having automatic eccentric clamps or on an Incline Plane Tester utilizing the principle of constant specimen-rate-of-load having suitable clamps for holding the specimen firmly. Distance between clamps to center shall be 10 inches and speed of pull shall be 12 inches per minute. Specimen of ligature shall be unknotted. The breaking strength shall be determined at two places on each five ligatures. The breaking strength shall be the average strength of the five ligatures.

4.4.2 Thread ply and number of threads. The thread ply and number of threads shall be determined with a pick glass or suitable magnifying glass and scale.

4.4.3 Sterility test. Test for sterility of the inner container shall be performed as follows: The outer container shall not be placed in any bactericidal solution. The outer container shall be opened, and the inner container removed in accordance with the manufacturer's instructions (see 3.7). The inner container shall then be submitted to sterility test in accordance with the current edition of the U.S.P.

4.4.4 Inner container leakage test. The inner containers shall be conditioned at ambient conditions for at least 4 hours prior to performing the test. A small quantity of water soluble detergent (approximately 4 grams per gallon of water or equivalent amount of wetting agent) may be added to the test water to remove surface entrapped air. The samples to be tested shall be submerged under water contained in a vacuum vessel from which they can be observed from the outside. The samples shall then be submitted to a vacuum of 8 inches of mercury (Hg) for 30 seconds.

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The above test shall be repeated after inverting the samples. While under vacuum, the containers shall be observed for bursting or leakage, indicated by a steady stream or recurring succession of bubbles from any surface or seam. Bubbles which may appear on the surface of the container are not to be construed as indication of failure.

4.4.5 Outer container leakage test. The outer container shall be completely submerged in water containing a suitable dye for 3 hours.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package (12 ligatures), as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in the following table shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	144 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable. Closure shall be adequate to prevent accidental opening under normal handling.

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5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

* 5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

* 5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

* 5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

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* 5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number, sterility lot (control) number and name of contractor shall be shown. In addition, the method of sterilization (G-Gas, R-Radiation, S-Steam, D-Dry Heat sterilization) shall be shown following the level of packaging and the date of packaging. Example: "A-9/69-G".

* 5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Sterility lot (control) number shall be shown. In addition, the method of sterilization (G-Gas, R-Radiation, S-Steam, D-Dry Heat sterilization) shall be shown following the level of packaging and the date of packaging. Example: "A-9/69-G".

Note: Final sterilization shall be on or after indicated packaging date.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification is intended to cover the types which are normally procured to meet Military requirements. It does not include all types, classes, grades and sizes as indicated by the title nor all types which are commercially available.

6.3 Transportation description. Transportation description applicable to this item is:

Tape, NOI, in packaged units of less than 24 continuous yards, in boxes.

Carload minimum weight: 20,000 pounds subject to Rule 34 of the Uniform Freight Classification.

Truckload minimum weight: 20,000 pounds per National Motor Freight Classification.

* 6.4 This specification covers the following item as listed in the Federal Supply Catalog.

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Federal Stock No.	Item Identification
6515-379-2100	LIGATURE, UMBILICAL, 18 inch, 12s

* 6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - MD
Navy - MS
Air Force - 03

Review activities:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1414

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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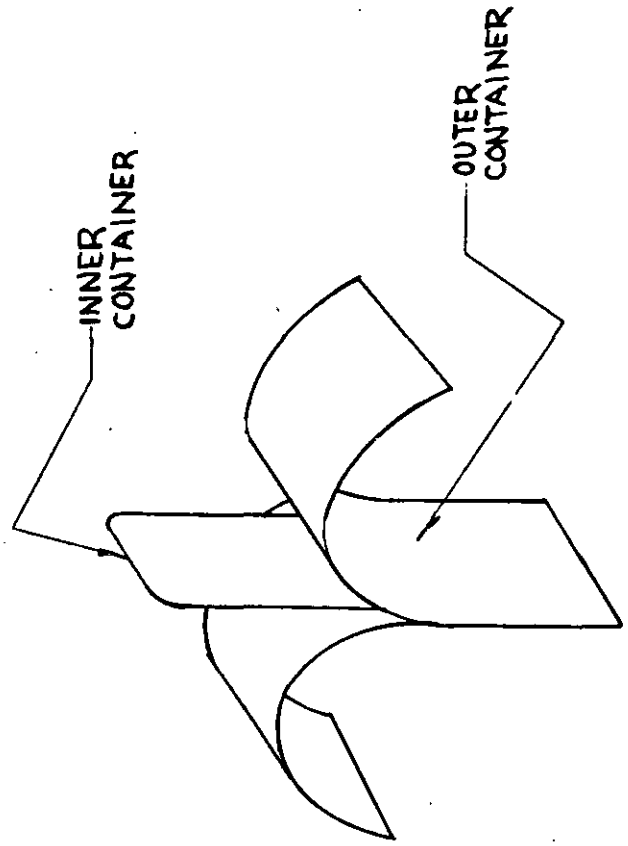


FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-L-2668D FSN:

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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